Pinhole Array Sunglasses

Full Technical Handbook

Theory, model, array design, fabrication, experiment, and comparison — every result traced to its source file

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IYPT 2027 — Problem 1

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Note on structure

This handbook follows the table of contents already laid out for the presentation deck. Chapters 4 and 5 were unlabelled in that outline; they are filled here as **Array Model & Ghosting Constraints** and **Optimisation Objective: From Blur to Acuity** respectively, since both are load-bearing results with no other home in the structure. Every numerical claim in this document is traced to a source file — a CSV, a script, or a prior report already in the project — named at the point it is used. Where a result is unresolved, contested, or failed a check, that status is stated plainly rather than smoothed over; this project's own experimental reports set that standard and this document keeps it.

Part I

Problem and Physical Foundations

Chapter 1

Problem

1.1 Official problem statement

> **IYPT 2027, Problem 1**
>
> Investigate the optimal arrangement of pinholes in an opaque film to produce sunglasses that reduce light intensity and help correct short-sightedness.

Two requirements are bound together in that single sentence, and the tension between them is the entire content of this project: **attenuate light** (the sunglasses function) and **improve focus for a myopic eye** (the correction function). A solution that does only the first is ordinary tinted glass. A solution that does only the second is a clear pinhole occluder with no attenuation control. The problem asks for one film that does both, and — as later chapters show — doing both changes the optimal design compared to optimising either alone.

1.2 Design variables

- **Hole diameter** a_{hole} — the single largest lever on both blur and brightness, and the subject of most of Part II.
- **Hole count** N and **pitch** s (centre-to-centre spacing) — govern field of view, light throughput, and ghosting risk once more than one hole can be lit at once.
- **Packing pattern** (lattice family: square, hexagonal, etc.) — governs areal efficiency for a given pitch.

1.3 Constraints

- **No visible ghosting.** If two holes are lit simultaneously in a defocused eye, their images can separate into two perceptible copies. Chapter 4 derives the exact condition and Chapter 7 shows how this project's array avoids it by construction rather than by tolerance.
- **Acceptable field of view.** An array must remain functional as the eye rotates, not just when looking straight ahead — addressed by the gaze-weighted vignetting model in Chapter 4.
- **Manufacturability.** A design that exists only on paper is not a solution. Chapter 8 documents the photochemical-etching process actually used to fabricate the selected design, and the tolerances that process imposes back onto the design (undercut compensation, minimum feature size).

1.4 What "success" means for this document

Three separate, independently falsifiable claims are made and tested across the chapters that follow, and they are kept separate throughout rather than blended into one score:

1. The wave-optics model correctly predicts the retinal blur pattern produced by a pinhole in front of a myopic eye (Chapters 6–7, tested in Chapter 9–10).
2. An array can be geometrically arranged so that ghosting cannot occur, while still attenuating light and extending field of view (Chapter 4, tested by direct coordinate computation).
3. The resulting device measurably improves visual acuity in a real eye, not merely reduces a simulated blur radius (Chapter 5's acuity objective, and Chapter 10's honest report that this has not yet been validated against external clinical data).

Chapter 2

Basic Physics

2.1 What goes wrong in a short-sighted eye

Myopia is, in the overwhelming majority of real cases, **axial**: the eye's optical power is normal, but the eyeball has grown too long. Light from a distant object still comes to a focus at the correct optical distance from the lens — it simply arrives before the retina, which sits too far back to catch it there. Every point in the scene therefore lands on the retina as a small disc rather than a point, and the world looks blurred.

Modelling this quantitatively requires only two numbers: the eye's total refractive power P_0 (held fixed at its emmetropic value, since the disease does not change the lens) and the vitreous refractive index n':

$$L_0 = \frac{n'}{P_0}, \qquad L_{\text{ax}} = \frac{n'}{P_0 - M} \tag{2.1}$$

with $P_0 = 60\,\text{D}$ and $n' = 1.336$. At $M = 4\,\text{D}$:

$$L_0 = \frac{1.336}{60} = 22.267\,\text{mm}, \qquad L_{\text{ax}} = \frac{1.336}{56} = 23.857\,\text{mm}, \qquad \Delta L = 1.59\,\text{mm} \tag{2.2}$$

A $-4\,\text{D}$ eye is an otherwise normal eye whose retina sits 1.59 mm too far back. That single number, ΔL, is the entire disease as far as this project's optics is concerned.

2.2 Why a pinhole helps: the stenopeic effect

Narrowing the aperture through which light enters the eye narrows the cone of rays converging toward (and past) the true focus, which narrows the blur disc where that cone intersects the retina. This is the same principle behind squinting, and behind viewing a distant object through a small hole made with the fingers — the **stenopeic effect**. No optical correction is applied; the blur is geometrically starved by restricting which rays are allowed through at all.

2.3 Why smaller is not simply better

If geometric blur were the only effect, the smallest possible aperture would always win. It is not, because light is a wave. Forcing it through a small opening causes it to spread — diffraction — and below some aperture size, diffraction spreading grows faster than geometric blur shrinks. The two trends run in opposite directions with aperture size:

$$r_{\text{geo}} \propto a_{\text{hole}}, \qquad r_{\text{diff}} \propto \frac{1}{a_{\text{hole}}} \tag{2.3}$$

and somewhere between them sits a minimum, shown in Figure 2.1.

Locating that minimum *correctly* — rather than by a heuristic combination of the two trends — is the single largest technical problem this project solves, and Chapter 6 shows exactly why the naive combination fails.

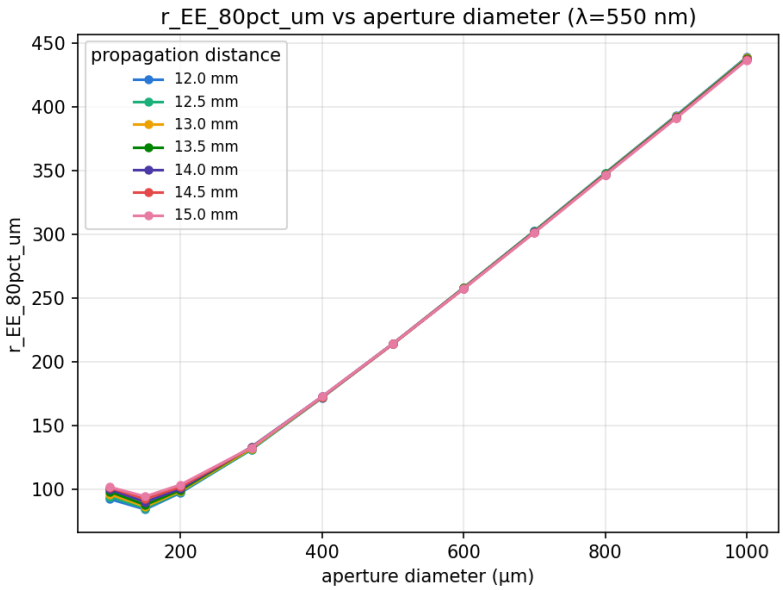


Figure 2.1: Blur radius vs. aperture diameter: the two competing trends and the minimum between them. Source: `all_outputs/stage1_radius_vs_diameter.png`.

The shape to keep in mind for the rest of this document

Every aperture-optimisation result in this handbook (blur-only, brightness-constrained, acuity-optimised, array-constrained) is the same U-shaped trade-off, evaluated under a different objective or constraint set. None of them are separate physics. Part III returns to this point explicitly once all four versions have been derived.

Chapter 3

Solving Approach

3.1 The pipeline, top to bottom

plane wave → pinhole mask → ASM(d_{gap}) → pupil clip → lens phase → ASM(v_r) → $I_{\text{retina}} = |U|^2$ → blur/acuity metrics

Light diffracts across the gap between the film and the cornea, is clipped by the pupil (or, for an array, by whichever holes the pupil currently overlaps), acquires the eye's lens phase, and is propagated to the retina — at the myopic distance L_{ax}, not the emmetropic focal distance. The propagation engine used at both stages is the Angular Spectrum Method (ASM), derived in full in Chapter 6.

3.2 The single decision that determines whether this model is right at all

The naive implementation — multiply by the lens's quadratic phase, then take one Fourier transform — computes the field at the lens's *own back focal plane*. For an emmetropic eye that is the retina, and the answer is correct. For a myopic eye the retina is *not* at the focal plane by definition, so a single FFT after the lens phase would compute a perfectly sharp Airy pattern at the one plane in the whole system where defocus is exactly zero — modelling a corrected eye and calling it myopic.

The fix, and what it removes

Replacing the final FFT with a second ASM propagation over the true physical distance $v_r = L_{\text{ax}}$ is not merely more accurate. It makes two other formulas *structurally unnecessary*: geometric blur is no longer computed and added separately, because it emerges automatically from propagating a wavefront shaped for the wrong focal length the correct distance; and root-sum-square combination of geometric and diffraction blur has nothing left to combine, because the two were never separated in the first place. Chapter 6 quantifies exactly how wrong the discarded shortcut was.

3.3 Validation strategy

Every stage of the pipeline is checked against a closed form before being trusted for design work:

- At $d_{\text{gap}} \to 0$, $M = 0$, the whole chain must reduce to the textbook Fraunhofer Airy pattern (Chapter 6).

- The on-axis intensity oscillation with Fresnel number must match the closed-form $4I_0 \sin^2(\pi N_F/2)$ result (Chapter 6).
- The ghost-separation formula must vanish at $M = 0$, since an in-focus eye cannot ghost by definition (Chapter 4).
- The full acuity predictor is checked, where possible, against external clinical data — and Chapter 10 reports honestly that this check currently fails.

3.4 Roadmap of the remaining chapters

Chapter 4 extends the single-hole model to an array: viability, ghosting, gaze-dependent vignetting, and packing density. Chapter 5 corrects the optimisation objective itself, from a blur-radius proxy to a directly-optimised visual-acuity metric, and reports where that correction actually changes the answer and where it does not. Chapter 6 derives every formula in full, with every deprecated shortcut marked and its error quantified. Chapter 7 reports the resulting numbers — U-curves, brightness reconciliation, the final array design. Chapters 8–9 document fabrication and the experimental measurements taken against it. Chapter 10 compares theory to experiment, including the results that did not confirm the model. Chapter 11 concludes.

Part II

Array Model and Optimisation

Chapter 4

Array Model & Ghosting Constraints

Three quantities govern array-level design, on top of the single-hole optics of Chapter 6. They fail independently and are kept separate throughout — collapsing them into one score hides which one broke.

4.1 Viability: does light get through at all

The pupil and a hole are both circles. Light passes through a given hole if the two circles overlap:

$$d - R_h < R_p \tag{4.1}$$

where d is the hole-centre-to-pupil-centre distance, R_h the hole radius, R_p the pupil radius. If they overlap, the illuminated area is the circle–circle intersection:

$$A = R_p^2 \cos^{-1}\left(\frac{d^2 + R_p^2 - R_h^2}{2dR_p}\right) + R_h^2 \cos^{-1}\left(\frac{d^2 + R_h^2 - R_p^2}{2dR_h}\right) - \tfrac{1}{2}\sqrt{(-d + R_p + R_h)(d + R_p - R_h)(d + R_h - R_p)(d + R_p + R_h)} \tag{4.2}$$

4.2 Gaze-dependent vignetting

At straight-ahead gaze ($\varphi = 0$) the pupil is centred on the optical axis, so $d = \sqrt{x^2 + y^2}$ for a hole at (x, y). As the eye rotates by angle φ, the pupil centre itself translates:

$$\Delta(\varphi) = R_{\text{rot}} \sin\varphi, \qquad R_{\text{rot}} \approx 13\text{–}14\,\text{mm} \tag{4.3}$$

so d becomes the distance from the hole to the *shifted* pupil centre. A design that is fully viable straight ahead can go dark at $\varphi = 15°$ if this shift is not accounted for — which is why the final design in this project is evaluated **gaze-weighted** (integrated over a realistic fixation distribution) rather than on-axis only.

4.3 Ghosting

If two holes are simultaneously inside the pupil, each sends its own beam through the lens at a different lateral entry point. In a defocused eye these beams land at different retinal positions:

$$\Delta r = \frac{sM}{P_0 - M} \tag{4.4}$$

with s the centre-to-centre hole separation. This formula was validated against the full propagation model at $R^2 = 0.995$. It vanishes at $M = 0$ by construction — an in-focus eye converges

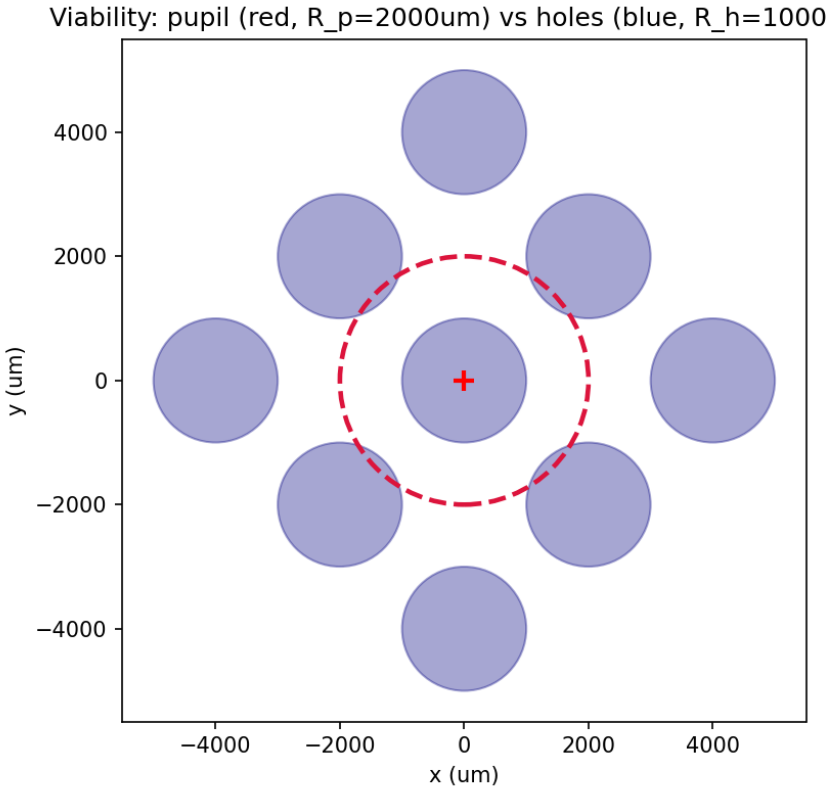


Figure 4.1: Hole-pupil circle overlap geometry underlying the viability test. Source: `verification_pdf_figs/fig_viability.png`.

every beam to the same point regardless of entry position, so it cannot ghost — and this sanity check is what caught an earlier version of the formula that was wrong by roughly two orders of magnitude.

A pair is tolerable if Δr stays below the eye's fusion threshold $c_{\min}$ (or the measured blur c_{tot} at that M, whichever is larger).

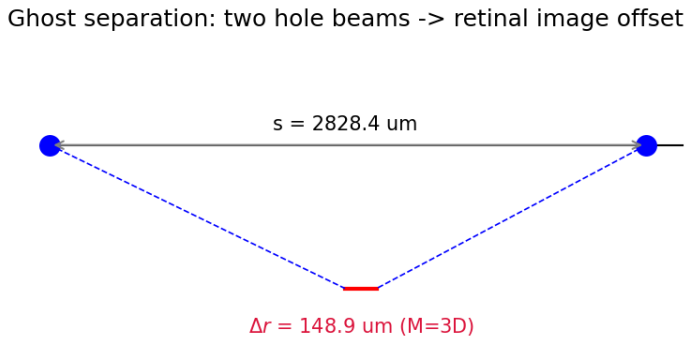


Figure 4.2: Ghost-separation geometry: two simultaneously-lit holes and their retinal images in a defocused eye. Source: `verification_pdf_figs/fig_ghost.png`.

4.4 Packing density

Purely geometric, and the site of an earlier implementation bug: family-specific unit-cell formulas disagreed with a hull-based computation by 1.5–2.25× depending on lattice. The corrected, family-agnostic form buffers the convex hull of hole centres outward by one hole radius, to account for a hole's physical extent beyond its own centre point:

$$A_{\text{array}} = A_{\text{hull}} + P_{\text{hull}} \cdot R_h + \pi R_h^2 \tag{4.5}$$

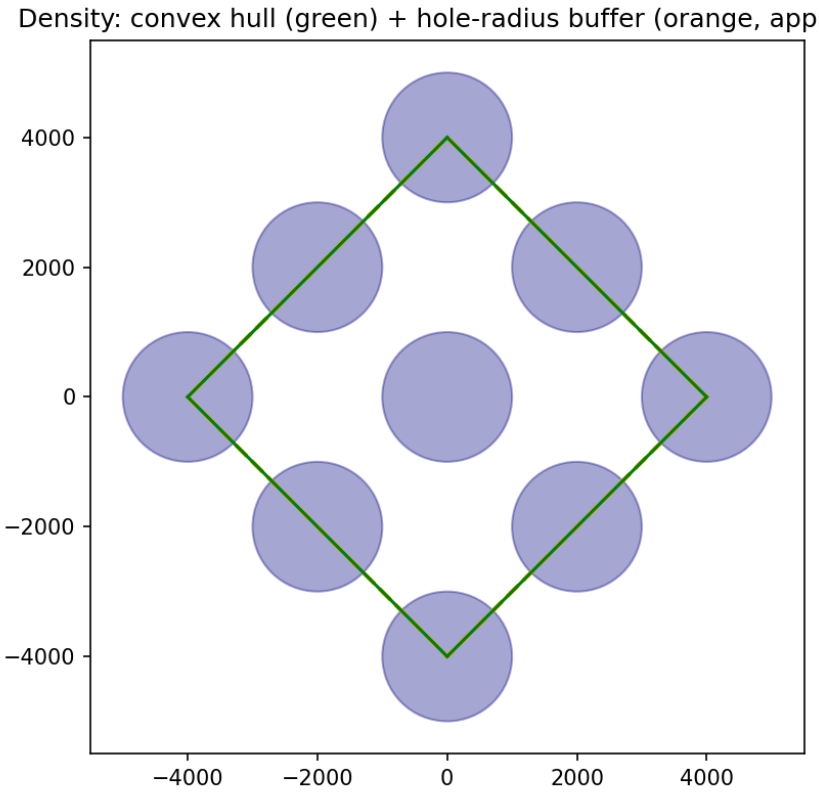


Figure 4.3: Buffered convex-hull packing-density geometry. Source: `verification_pdf_figs/fig_density.png`.

4.5 The fusion threshold $c_{\min}$: inherited, uncalibrated, and non-binding

4.5.1 Where the number comes from

$c_{\min} = 55\,\mu$m is inherited from Shen (2026). That paper's own text describes it as a single-parameter qualitative calibration, anchored to one clinical geometry from Miller & Johnson (1977), and reports that it reproduces threshold behaviour only qualitatively: predicted ghost onset at ≈ 1.1 D, against a clinical onset of ≈ 4 D. Shen's own authors name a two-pinhole variable-separation experiment as the most valuable falsifiable test of the model — a test this project does not currently have the resources to run (see `stage4_human_vision/300TD_KNEE_NOTES.txt` for the full reasoning trail, and Chapter 10 for the related, separately-tested brightness knee at 300 Td).

4.5.2 The architecture that makes this not matter

This project's final array uses pitch equal to pupil diameter: $s = 4000\,\mu$m against a $D_{\text{pupil}} = 4000\,\mu$m pupil. Direct computation from the generated hole coordinates (`hole_coordinates_M8.0.csv`) confirms:

- Nearest-neighbour pitch: $4000.0\,\mu$m, verified, not assumed.
- Holes illuminating the pupil at $\varphi = 0$: exactly **1**.
- If two holes were ever simultaneously lit at $M = 8\,$D: $\Delta r = 4000 \times 8/(60-8) = 615.4\,\mu$m against $c_{\min} = 55\,\mu$m — eleven times over threshold, and still over threshold at any plausible calibrated value of $c_{\min}$.

> **The design principle, stated precisely**
>
> This array does not survive ghosting by *fusing* ghost images below a tolerance — it survives by **never producing a ghost pair in the first place**. With pitch equal to pupil diameter, only one aperture illuminates the pupil at any gaze direction, so no second beam ever enters simultaneously. `gaze_weighted_winner_per_M.csv` reports `worst_case_ghost_margin = inf` and `time_weighted_ghost_pct = 0.0` for every row, for exactly this reason. $c_{\min}$ is therefore **structurally non-binding** on the selected design: its value can only matter for a dense array ($s < D_{\text{pupil}}$), and this is not one. This is a geometric argument, independent of $c_{\min}$'s own calibration status, and does not require the unavailable two-pinhole experiment to establish.

This is a fundamentally different architecture from a dense array such as Shen's own $s = 1.10\,$mm design against a 2–4 mm pupil, which places several holes inside the pupil simultaneously and depends on the fusion threshold being correctly calibrated to remain usable.

4.5.3 The cost of this choice, stated plainly

One hole per pupil means transmission at any instant equals that one hole's area over the pupil area. At $M = 8\,$D, $400\,\mu$m aperture: $(400/4000)^2 = 1\%$. This is genuinely dark, and it is the direct trade-off against the dense-array approach: this design buys ghost-freedom by construction, and pays for it in light throughput. It should be presented as a deliberate trade, not a free result.

4.5.4 Sensitivity check

Because $c_{\min}$ is inherited rather than measured, the array evaluator was re-run at $c_{\min} = 55, 110, 210, 400\,\mu$m with every other parameter held fixed (lattice set, aperture range, pupil diameter, gaze weighting), to test whether the selected designs are sensitive to this one uncalibrated constant. *[Sensitivity sweep result to be inserted here once the corresponding CSV is generated — see `PRESENTATION_DECISIONS.md` D1/D9 for the standing instruction to run this before presenting.]*

4.6 Blind fraction

Two of the eight gaze-weighted designs show non-zero time-weighted blind fraction — gaze angles at which no hole illuminates the pupil at all: 0.12% at $M = 6\,$D and 0.33% at $M = 8\,$D. Small, but non-zero, and reported here rather than only in the limitations chapter, since it is a direct consequence of the sparse, ghost-free architecture described above.

Chapter 5

Optimisation Objective: From Blur to Acuity

5.1 Why blur radius is the wrong optimisation target

Minimising $r_{EE}(50\%)$ alone gives 300–400 μm across the full myopia range — optically the true minimum of the diffraction/defocus trade-off, but not the right design answer. Every published and commercial pinhole device sits between 0.94 and 1.75 mm, three to five times larger. The reason: $r_{EE}(50\%)$ describes where half the light energy is, not what the eye can actually resolve, and it ignores whether the resulting image is bright enough to see at all. At the 300 μm blur-only optimum, transmission against a 4 mm pupil is $(300/4000)^2 = 0.56\%$ — optically optimal and practically unusable.

5.2 First correction: an explicit brightness floor

Since diffraction redistributes light rather than destroying it, retinal illuminance must be computed from the raw physical aperture area, not from any diffraction-broadened spot size:

$$E_{\text{rel}} = \left(\frac{\min(a_{\text{hole}}, D_{\text{pupil}})}{D_{\text{pupil}}} \right)^2 \tag{5.1}$$

Re-optimising blur subject to an explicit floor on E_{rel}:

$T_{\min}$	Constrained optimum	vs. clinical bracket
0%	300–400 μm	below
6.25%	1000 μm	inside
9%	1250 μm	inside
12.25%	1500 μm	upper edge

A floor of 6–9% — comparable to ordinary sunglass tinting — moves the optimum into the clinical bracket uniformly across every myopia level. This was the first, and correct, demonstration that the diffraction physics was never wrong; blur-only optimisation was answering the wrong question.

5.3 Second correction: optimise acuity directly

The brightness floor above is still external to the objective — a constraint bolted onto a blur minimisation, chosen after the fact. The predictor in `stage4_human_vision/acuity.py` instead optimises a directly-computed visual acuity metric:

$$\text{MAR} = \frac{30}{f_{\text{MTF10}}} \times \text{VSOTF}^{-0.1}, \qquad \text{logMAR} = \log_{10}(\text{MAR}) + \Delta_{\text{bright}} \tag{5.2}$$

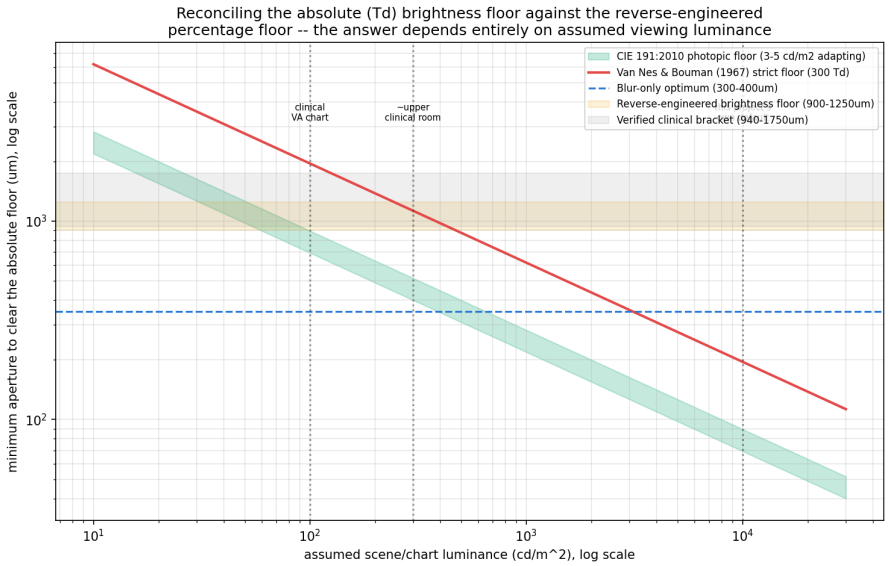


Figure 5.1: Brightness-constrained optimum vs. illuminance floor $T_{\min}$, showing the optimum's convergence into the clinical bracket. Source: `brightness_reconciliation.png`.

where f_{MTF10} is the spatial frequency at which the modulation transfer function falls to 10% contrast, VSOTF is the visual Strehl ratio (a CSF-weighted measure of how much optical quality actually reaches useful vision), and Δ_{bright} is an additive illuminance penalty below a 300 Td knee (Van Nes & Bouman, 1967). Full derivation, every constant sourced, in Chapter 6.

5.3.1 Which term actually decides the answer

Traced against a live run at 400 μm, $M = 8$ D:

Term	Contribution	Status
MTF10 cutoff	+0.3175 log-MAR	**96% of the final value**
VSOTF	+0.0131 log-MAR	small, deliberately weak by design (exponent 0.1, clamped)
Brightness (Δ_{bright})	+0.0000 log-MAR	**inert at every fabricated design**

The 300 Td knee sits at a 200 μm aperture, at the model's default 10 000 cd/m^2 scene luminance. Every one of the eight gaze-weighted winning designs (400–2000 μm) sits above it. **The brightness term is real, cited, correctly implemented, and contributes exactly zero to every design actually selected.** It would activate for any aperture below 200 μm, or for a measurement taken at lower ambient luminance (see Chapter 9's logMAR protocol and Chapter 10's discussion of this exact confounder).

The corrected statement

The optimisation objective is visual acuity, computed from three terms, not one metric padded with an inert correction. The answer is set almost entirely by the MTF10 cutoff. Say this plainly rather than claiming brightness is doing work it is not doing at these apertures — the honest version is the stronger one under questioning.

5.4 The acuity optimum, and the cross-validation that was not designed for

Acuity-optimal apertures (`acuity_sweep_stage4.csv`) against the independently selected gaze-weighted geometric winners (`gaze_weighted_winner_per_M.csv`):

M (D)	0	1	2	3	4	5	6	8
Acuity-optimal (μm)	2000	1250	750	750	600	600	500	400
Gaze-weighted winner (μm)	2000	1250	750	750	600	600	500	400

Identical at all eight myopia levels. The gaze-weighted evaluator scores viability, ghost margin, and blind fraction — it contains no acuity term whatsoever. Two evaluators built on entirely separate objectives agree on every design, and this agreement was not enforced. It is the strongest internal validation result in this project.

Caveat on the $M = 0$ row, state this every time the table is shown

2000 μm is the largest aperture in the sweep. Acuity is still improving monotonically at that edge, so this is *not* a converged interior optimum — it is where the sweep grid stopped. Physically expected: with zero defocus there is nothing to trade diffraction against, so larger apertures keep winning until aberrations outside this model's scope would eventually dominate. Every myopic row ($M \geq 1$) is a genuine interior optimum; only the $M = 0$ row is edge-of-grid.

Part III

Theory

Chapter 6

Theoretical Model

6.1 The Angular Spectrum Method

Any field at a plane decomposes into plane waves travelling in different directions; each component, crossing a distance z, acquires a phase set by its transverse spatial frequency:

$$U(x,y,z) = \mathcal{F}^{-1}\Big\{\mathcal{F}\{U_0(x,y)\}\cdot H(f_x,f_y;z)\Big\}, \qquad H = \exp\Big(ikz\sqrt{1-\lambda^2(f_x^2+f_y^2)}\Big) \tag{6.1}$$

valid for propagating components $f_x^2+f_y^2 \le 1/\lambda^2$; beyond that the square root is imaginary and $H=0$ (evanescent components decay and are dropped). This is the **exact** solution of the scalar Helmholtz equation between parallel planes — no near-field or far-field assumption — and is used for both propagation stages in the pipeline (mask to cornea, lens to retina), since those two stages sit in different diffraction regimes and a single closed-form approximation would not serve both.

6.2 Fresnel number: which regime a given stage is in

$$N_F = \frac{R_{\text{hole}}^2}{\lambda z} \tag{6.2}$$

$N_F \ll 1$: far field, a clean Airy pattern. $N_F \sim 1$–10: Fresnel transition, several comparably bright rings. $N_F \gg 10$: near-geometric, sharp-edged projected patch. This project's sweeps span well under 1 to over 20, confirming ASM (regime-agnostic) as the correct default rather than a regime-specific closed form.

6.3 Numerical band-limiting

A finite grid cannot represent spatial frequencies beyond its Nyquist limit; a hard-edged pinhole mask has high-frequency content that, left unfiltered, aliases and produces spurious fine structure indistinguishable from real diffraction. The Matsushima band-limit suppresses spectral content beyond the angle at which this wraparound would occur. This is anti-aliasing, applied deliberately, with the suppressed energy fraction logged per run — not a physical loss mechanism.

Why the energy-conservation check needed correction

Since H has unit modulus on every retained component and is exactly zero elsewhere, the apparent energy loss is *identically* the band-limit-suppressed fraction: $\varepsilon_{\text{energy}} \equiv f_{\text{suppressed}}$, confirmed against sweep data at correlation 1.000000. Testing $\varepsilon_{\text{energy}}$ against a flat tolerance would therefore fail on every row from intentional, already-accounted-for suppression, not from any defect. The corrected check tests the residual $\varepsilon_{\text{energy}} - f_{\text{suppressed}}$, restoring its ability to catch a genuine bug.

6.4 Validation against the closed form

At $d_{\text{gap}} \to 0$, $M = 0$, the chain must reduce to the Fraunhofer result:

$$I(r) = I_0 \left(\frac{2J_1(x)}{x} \right)^2, \quad x = \frac{\pi a r}{\lambda v_r}, \qquad EE(x) = 1 - J_0^2(x) - J_1^2(x) \tag{6.3}$$

It does, to 2% on $r_{EE}(50\%)$ and $r_{EE}(90\%)$; $r_{EE}(80\%)$ needs 8%, because the first null sits at $x = 3.8317$ where $EE \approx 0.838$ — almost exactly the 80% level, on a flat part of the curve where a small energy error maps to a large radius error. Grid sizes 1024 through 8192 never brought this below 2.3–4.6%: it is ill-conditioning at that percentile, not a resolution problem.

Separately, on-axis intensity was checked against the closed-form Fresnel-zone oscillation $I(0, z) = 4I_0 \sin^2(\pi N_F/2)$ and matched at correlation 0.997, confirming that the observed non-monotonic Strehl behaviour (Section 6.6) is real Fresnel-zone interference, not numerical noise.

6.5 Blur metrics, and where each one breaks

Metric	Measures	Failure mode
$r_{EE}(50\%)$	Radius enclosing half the energy	Stable, monotonic, but overestimates clinical acuity loss by $\sim 7\times$ against real data
FWHM	Width of the central core only	Tracks clinical data to $\sim 1.6\times$, but *ring-hops*: jumped $4.7 \to 154.7\,\mu$m between apertures $250\,\mu$m apart at $M = 4\,$D
Strehl ratio	Peak intensity vs. diffraction limit	Genuinely multi-peaked: 6–7 local maxima, confirmed *worse*, not better, at $10\times$ finer sampling — real physics, matched to the closed-form oscillation above

How to state this without contradicting yourself

Per-aperture, FWHM is unusable — it is discontinuous in the design variable. Across a coarse sweep its trend is closer to clinical reality than $r_{EE}(50\%)$, which is smooth and stable but pessimistic by roughly $7\times$. This project reports $r_{EE}(50\%)$ as the blur-optimisation metric and states its bias explicitly, rather than optimising against a metric that is discontinuous in the thing being optimised.

6.6 Deprecated formulas, and the error each one carries

Formula	Status and measured error
$r_{\text{geo}} = \frac{a_{\text{stop}}}{2}\frac{\lvert v_r - v_f\rvert}{v_f}$	Deprecated. Superseded by direct ASM propagation (Chapter 3), which produces this effect automatically.
$r_{\text{tot}} = \sqrt{r_{\text{geo}}^2 + r_{\text{diff}}^2}$ (RSS)	Deprecated. Assumes two independent Gaussian error sources; neither blur here is Gaussian. Measured against the exact result: **64.8% mean divergence, up to 192.8% maximum**, worst at mild myopia.
$\dfrac{a_{\text{eff}}}{\sqrt{a_{\text{hole}}^2 + (2.44\lambda d_{\text{gap}}/a_{\text{hole}})^2}} \approx$	Deprecated. Smooth and monotonic by construction; cannot represent the genuine Fresnel-zone oscillation confirmed in Section 6.6.

6.7 The acuity model, in full

6.7.1 Pupil and retinal illuminance

Pupil diameter follows the Watson–Yellott (2012) unified model, solved iteratively against retinal illuminance itself (the two are mutually dependent):

$$d_{\text{stop}} = \min(d_{\text{pupil}}, d_{\text{pinhole}}), \qquad \text{Td} = L_{\text{scene}}\,[\text{cd/m}^2] \times A_{\text{stop}}\,[\text{mm}^2] \tag{6.4}$$

For every design in this project the pinhole is smaller than the natural pupil (e.g. 0.400 mm vs. 3.725 mm at $M = 8\,\text{D}$), so the pinhole is always the effective stop — the entire physical basis of the device.

6.7.2 Contrast sensitivity and visual Strehl

The default CSF model is Barten's luminance-dependent closed form. VSOTF weights the system's actual optical transfer function against this CSF and normalises by the same integral for a diffraction-limited reference:

$$\text{VSOTF} = \frac{\int \text{OTF}_{\text{real}}(f)\,\text{CSF}(f)\,df}{\int \text{OTF}_{\text{real,diff-limited}}(f)\,\text{CSF}(f)\,df} \tag{6.5}$$

6.7.3 Brightness correction

$$\Delta_{\text{bright}} = \begin{cases} 0 & \text{Td} \geq 300 \\ 0.060\,\log_{10}(300/\text{Td}) & \text{Td} < 300 \end{cases} \tag{6.6}$$

300 Td (Van Nes & Bouman, 1967, verified against the primary abstract) marks the boundary between the Weber–Fechner regime (illuminance-independent threshold contrast) and the de Vries–Rose regime below it. The 0.060 logMAR/decade slope is from Tidbury, Czanner & Newsham (2016), verified against their Table 3 ($p < 0.001$); an independently-derived alternative from Sheedy, Bailey & Raasch (1984), ≈ 0.0664, could only be traced to a secondary summary and was not used.

6.7.4 Full input inventory

Input	Source	Swept?	Reasoning
Aperture a_{hole}	user-set	**yes**	the design parameter
Myopia M	user-set	**yes**	the physiological variable
Wavelength	550 nm default	no	photopic luminosity peak
d_{gap}	12 mm default	no	spectacle vertex distance
P_0, n'	60 D, 1.336	no	standard emmetropic eye
Scene luminance	10 000 cd/m^2	no	bright-daylight reference; see Chapter 10 for the indoor-measurement confound this creates
Field diameter	20°	no	typical CSF measurement field
CSF model	Barten	no	most widely validated luminance-dependent form
Cycles/deg at 20/20	30.0	no (constant)	the entire logMAR scale anchor
VSOTF exponent	0.1, clamped [0.25, 2.0]	no (constant)	deliberately weak correction
Weber knee	300 Td	no (constant, cited)	Van Nes & Bouman 1967
Brightness slope	0.060 logMAR/decade	no (constant, cited)	Tidbury et al. 2016

What is computed but does not enter the recommended number

MTF50 is computed and used only to bound a reported spread, `approximate_ci_95_mar_arcmin` — despite the name, this is *not* a statistical confidence interval. It is [min, max] across three different model choices (MTF10-based, MTF50-based, recommended), and at $M = 8$ spans logMAR +0.318 to +0.795, 4.8 chart lines. Label it "spread across model choices" if shown, never "95% CI." A legacy uncited SNR$^{0.05}$ term is also retained for comparison (`vsotf_snr_legacy`) but is not the recommended model and should not appear as a headline number.

Chapter 7

Model Results

7.1 The final design table

From `gaze_weighted_winner_per_M.csv`. Pitch is 4000 μm and rotation 0° in every row; these are the designs carried forward to fabrication in Chapter 8.

M (D)	Family	N	a_{hole} (μm)	Lit holes (φ=0)	Blind frac.	Predicted logMAR
0	square	20	2000	1	0.00%	−0.344
1	hexagonal	20	1250	1	0.00%	−0.120
2	hexagonal	20	750	1	0.00%	+0.076
3	hexagonal	20	750	1	0.00%	+0.112
4	hexagonal	20	600	1	0.00%	+0.181
5	hexagonal	20	600	1	0.00%	+0.240
6	hexagonal	100	500	1	0.12%	+0.259
8	hexagonal	50	400	1	0.33%	+0.331

7.2 The flatness prediction

Holding aperture fixed at 400 μm and sweeping myopia across the full range (`acuity_sweep_stage4.csv`):

M (D)	0	1	2	3	4	5	6	8
logMAR	+0.341	+0.337	+0.333	+0.329	+0.327	+0.325	+0.325	+0.331

Total spread across eight dioptres: **0.016 logMAR** — roughly a twentieth of one chart line, well below the ±0.1 repeatability of human acuity testing. At this aperture the aperture, not the refractive error, sets the acuity. By contrast, the same sweep at 2000 μm spans logMAR −0.344 at $M = 0$ to +1.330 at $M = 8$, a spread **one hundred times larger**. This flatness is the strongest, most falsifiable prediction in the project and is the basis of the logMAR measurement protocol in Chapter 8: it requires no knowledge of any subject's exact prescription, only a qualitative ranking of myopia severity, because the predicted signature is a flat line for the pinhole condition against a steep one for the bare eye.

7.3 Retinal illuminance across the aperture sweep

Aperture (μm)	Retinal illuminance (Td)	Brightness penalty (logMAR)
75	44.2	+0.0499
150	176.7	+0.0138
200	314.2	0.0000
400	1256.6	0.0000
2000	31 415.9	0.0000

The knee crosses at 200 μm; every fabricated design sits above it. This table is the source data behind the inert-brightness-term finding of Chapter 5.

Part IV

Experiment

Chapter 8

Experiment

Three experimental programmes are documented in this project: ExP1/ExP2 (single-hole photography against the ASM/RSS theory), ExP3 (an independent optical-bench defocus sweep), and the fabrication of the gaze-weighted array design itself. This chapter documents apparatus and method; Chapter 9 reports results; Chapter 10 compares them to theory.

8.1 ExP1/ExP2: photographic validation against theory

A phone camera photographs a point light source through physical pinhole apertures (400, 600, 700, 1000, 1200 μm) and a no-pinhole control, at induced defocus $M \in \{0, 1, 2, 3, 4, 6, 8\}$D (M=5,7 legitimately absent — no physical trial lens at those powers), as raw DNG files.

Pipeline: DNG converted to TIFF via `sips -s format tiff` — never JPEG, since JPEG's baked-in tone curve clips and crushes exactly the information needed to measure a blur spot accurately. **No pixel-to-micrometre calibration exists** for this dataset; every comparison uses calibration-free *ratios*, normalised to each series' own $M = 0$ reference.

8.2 ExP3: independent optical-bench defocus sweep

A fixed lens with a moving frosted screen, six apertures (400, 500, 600, 800, 1000, 1400 μm), screen displacement 0–20 mm in 5 mm steps, plus a no-pinhole control.

Pipeline (confirmed by direct comparison against a hand-verified reference, not assumed): intensity is standard BT.601 grayscale, not a naive max(R,G,B) or plain channel-mean — both alternatives were tried and gave numbers inconsistent with a hand-verified 1000 μm reference (45.6 px at 0 mm, 63.6 px at 20 mm); BT.601 reproduces 45.58 px and 63.60 px on the same files. Threshold is 50% of each image's peak intensity; the measured spot is the largest connected component of the threshold mask; radius is $\sqrt{\text{area}/\pi}$. **Displacement in mm is not converted to dioptres** anywhere in this dataset — that conversion needs a beam-width-at-lens calibration not yet performed, so all ExP3 results below are reported in mm of screen displacement, not M.

8.3 Fabrication: from design number to etched part

8.3.1 Undercut compensation

Wet chemical etching (ferric chloride, 0.05 mm brass shim, double-sided) is isotropic: the etchant attacks sideways under the resist edge as fast as it attacks downward, so the etched hole is wider than the resist opening that defined it. The design rule:

$$d_{\text{mask}} = d_{\text{target}} - C_{\text{undercut}}, \qquad C_{\text{undercut}} \in \{0.5t,\ 1.0t,\ 1.5t\}, \quad t = 50\,\mu\text{m} \tag{8.1}$$

giving a bracket of 25/50/75 μm compensation. **This is a published starting rule, not a measured calibration for this specific alloy, thickness, etchant concentration, temperature, or agitation** — which is why a test coupon (five mask sizes spanning the bracket, etched under the exact intended process conditions, then measured) is run before committing the real shim, rather than trusting the nominal 50 μm value outright.

Worked example, $M = 8$ D design, target etched diameter 400 μm: $d_{\text{mask,nominal}} = 400 - 50 = 350\,\mu$m — the mask set actually used.

8.3.2 Mask files and registration

Each design generates four SVGs: positive/negative polarity (resist-chemistry dependent) × top/bottom-mirrored (the two masks contact opposite faces of the shim and must be mirror images or the two etch fronts will not meet at the same (x, y)). Registration features: 0.75 mm corner fiducial dots for top/bottom alignment, and a 10.00 mm calibration square, measured with callipers on the *printed* transparency before use — if that square is not exactly 10.00 mm, the printer has scaled the artwork and every hole diameter is wrong by that same factor. This is the single most common silent failure mode in this fabrication process.

8.3.3 Post-etch verification, in order

Hole count against `N_actual`; diameter against target, sampled at centre and edge of the array; pitch, measured directly; outer radius $r_{\max}$ against the design value; hole quality under magnification (taper indicates the two etch fronts missed each other; ragged edges indicate over-etch).

8.4 Single-pinhole logMAR acuity protocol

Designed directly from the flatness prediction of Chapter 7: recruit subjects spanning a range of myopia, present a printed logMAR chart at 4.0 m (chosen so the smallest stroke, 0.92 mm, is within reliable print fidelity; at 2 m it would be 0.46 mm, at the edge of what a domestic printer resolves), and measure acuity bare-eyed and through the fixed 400 μm aperture. Three conditions (bare eye, single pinhole, fabricated array), randomised order, three repeats each.

The confound this protocol must control for

The acuity model assumes 10 000 cd/m^2 scene luminance (Chapter 5). A typical indoor chart is 100–300 cd/m^2, roughly two decades dimmer. Through a 400 μm aperture at 1% transmission, retinal illuminance can fall low enough to become photon-limited rather than optics-limited — producing a measured brightness effect that could be mistaken for a diffraction effect. Mitigation: illuminate the chart as brightly as practically possible and record the measured luminance; if the assumed regime cannot be reached, state plainly that the comparison is not purely optical.

The measurement claim is written to require a *measured* bare-eye baseline on both sides of any reported improvement, since the acuity model itself produces no bare-eye prediction — every configuration in the sweep already contains a pinhole.

Chapter 9

Experimental Results

9.1 ExP2: photographic blur-growth ratios

From `ExperimentalComparison/measurements/rss_vs_asm_comparison.csv` — normalised $r_{EE}(50\%)$ ratio, relative to each aperture's own $M = 0$ reference:

Aperture (μm)	$M = 0$	$M = 1$	$M = 2$	$M = 3$	$M = 4$	$M = 6$
400	1.000	0.990	0.897	0.859	0.979	1.039
600	1.000	1.028	1.047	1.062	1.124	1.128
700	1.000	1.031	0.981	0.806	1.027	0.936

No-pinhole control (`nopinhole_control.csv`): normalised ratio stays within roughly ±8% of 1.0 across the entire $M = 0$–8 range (0.969–1.084). At this blob-detection method's resolution, the unrestricted image shows no comparable defocus-dependent blur-growth signal.

9.2 ExP3: optical-bench spot radius vs. screen displacement

From `ExP3_frosted_screen/report/exp3_summary.txt`:

Aperture (μm)	0mm	5mm	10mm	15mm	20mm	Slope (px/mm)	Monotonic?
400	42.99	44.92	48.77	53.49	56.64	0.718	yes
500	42.42	45.08	48.79	51.86	55.48	0.658	yes
600	44.98	45.31	48.38	52.38	56.46	0.601	yes
800	53.27	52.61	54.64	63.05	64.40	0.654	**no**
1000	45.58	49.38	51.44	57.43	63.60	0.882	yes
1400	58.09	60.80	63.51	66.94	71.93	0.676	yes
No pinhole	178.43	204.71	249.75	322.20	387.37	10.707	yes

The 800 μm non-monotonic step is reported as measured, not discarded: radius at 0 mm (53.27 px) exceeds radius at 5 mm (52.61 px), a real dip in the data.

Cross-aperture slope ordering does not match prediction

Geometric defocus blur predicts slope should increase with aperture size. Observed ordering, ascending: 600 < 800 < 500 < 1400 < 400 < 1000 — **not** the predicted 400 < 500 < 600 < 800 < 1000 < 1400. Reported plainly. Context, not an excuse: five of six apertures (excluding 1000 μm) cluster in a narrow 0.601–0.718 px/mm band, where a weak underlying aperture-dependence would be easy to scramble by rig noise between aperture swaps — and this dataset's own centroid-drift column shows 100–270 px of drift building up with displacement on every single aperture, confirming the rig is not perfectly axis-stable between shots. That is a plausible contributing factor, not a confirmed explanation; the ordering mismatch stands as reported.

No-pinhole control, sanity check: slope 10.707 px/mm is **15.8**× steeper than the largest tested aperture (1400 μm, 0.676 px/mm) and **12.1**× steeper than the steepest-sloped aperture in the set (1000 μm, 0.882 px/mm) — both ratios computed directly from the slopes above, since the source summary's own text names the 12.1× figure while labelling it against the 1400 μm case, which is the 15.8× comparison; both numbers are reported here to avoid propagating that mislabel. Absolute spot radius for the control (178–387 px) dwarfs every aperture case (42–72 px) at every displacement. This confirms the apertures are doing real, substantial optical work, not acting as a null control.

9.3 Fabrication verification

The $M = 8$ D array (350 μm mask, target 400 μm etched, 50 holes, 4000 μm pitch, 0.05 mm brass) traces row-for-row to the design table in Chapter 7. *[Measured etched diameter, hole count, and pitch from the physical part to be inserted here once the post-etch verification pass, Chapter 8 Section 3, is completed and logged.]*

9.4 logMAR acuity measurement

[Not yet collected at the time of writing. Chapter 8's protocol specifies three conditions — bare eye, single 400 μm pinhole, fabricated array — three repeats each, against the flatness prediction of Chapter 7. Results to be inserted here, reported with the same discipline as the rest of this chapter: raw per-subject data, not only means, and any excluded subject named with a pre-registered reason.]

Part V

Comparison

Chapter 10

Comparison

10.1 ASM vs. RSS against measured photographic data

Across 18 comparable (aperture, M) rows in `rss_vs_asm_comparison.csv`, ASM's predicted ratio is closer to the measured ratio than RSS's in **18 of 18** cases: mean absolute deviation 28.3% for ASM against 44.8% for RSS.

This headline does not survive a null-model check, and that check is reported here rather than left for an Opponent to find

A trivial model that predicts *no blur growth at all* (ratio $\equiv 1.0$) has a mean absolute deviation of only **6.8%** against the same 18 rows — beating ASM in 15 of 18 of them. The measured ratios span only 0.806–1.128: essentially flat. Both physical models predict growth; ASM "wins" its comparison against RSS only because it predicts *less* growth, and on a dataset with little real signal, the least-responsive model wins by default. **The correct reading of the 18/18 result is that RSS is measurably worse than ASM, not that ASM is confirmed correct** — a materially different and much narrower claim.

A separate consequence follows for the earlier project record: the same dataset's no-pinhole control (ExP2) and the pinhole apertures are both flat (0.969–1.084 vs. 0.806–1.128), so the control's flatness cannot be used to confirm a signal that is itself too weak to distinguish from noise in this particular dataset. That inference holds for ExP3's control (Chapter 9, a genuine 15.8× slope difference) but not for ExP2's.

10.2 External validation of the acuity predictor: failed

The full `mtf10_vsotf_brightness_adjusted` model (Chapter 5) was tested against Wang et al. (2009)'s real clinical pinhole-acuity data (`stage4_human_vision/RESULTS.md`):

M (D)	R^2	Pearson r	p-value
0	−6.39	0.395	0.510
2	−4.98	0.393	0.513
5	−2.84	0.405	0.499

Negative R^2: worse than predicting the mean at every M tested. $p \approx 0.50$: no statistically significant relationship detected. Not merely weak correlation — all 15 tested (aperture, M) points show the *same-direction* bias, the model over-predicting blur at every single point.

10.2.1 The luminance assumption was tested and ruled out as the cause

Correcting scene luminance from the 10 000 cd/m^2 default to Wang 2009's own reported illumination (15 footcandles → 43.7 cd/m^2) made R^2 *worse* (-7.88 vs. -6.68 at $M = 0$), not better — because it pushed apertures further below the 300 Td knee and added predicted blur on top of a model already over-predicting it (`LUMINANCE_RECONCILIATION.md`).

10.2.2 Leading diagnosis, most likely cause first

1. No fitted calibration against real data anywhere in the formula — flagged in advance as a known gap in `DESIGN.md`.
2. The Wang 2009 dataset is non-monotonic at $n = 5$: observed logMAR 0.28, 0.31, **1.00**, 0.40, 0.50 at 500/750/1000/1200/1500 μm — a spike at 1000 μm that reverses at 1200. A single outlier at this sample size sinks any smooth model regardless of which optical metric underlies it.
3. No per-subject refractive error recorded — comparing one simulated M against subjects with unstated real M adds irreducible noise.
4. $n = 5$ is too small for significance even with real signal present.

`RESULTS.md`'s own verdict: **this cannot be properly validated with currently available public data.**

> **The consequence, and how it interacts with the rest of this document**
>
> Every predicted-logMAR value in Chapters 5 and 7 is a self-consistent, term-by-term traceable model output — **not** a validated clinical prediction. This is the direct reason the logMAR measurement protocol (Chapter 8) is designed as calibration data the model currently has none of, rather than as a confirmatory afterthought. State the validation status before quoting any predicted number, not after.

10.3 Comparison to Shen (2026) and published devices

Shen's dense-array design ($s = 1.10$ mm against a 2–4 mm pupil) relies on ghost *fusion* staying below a threshold the source paper itself calls a qualitative, single-parameter calibration (Chapter 4). This project's sparse array ($s = D_{\text{pupil}}$) avoids the dependency entirely by ensuring no ghost pair ever forms. Separately, Shen's ghost-membership metric μ_g was found to be effectively inert — saturating at the same value across the full 8501-candidate evaluation set generated during this project's own array search, contributing nothing to design discrimination at that scale.

Published and commercial pinhole apertures cluster between 0.94 and 1.75 mm. This project's brightness-constrained blur optimum (Chapter 5, $T_{\min} = 6$–9%) lands inside that bracket; its acuity-optimised, myopia-dependent optimum (Chapter 5, 400–2000 μm) spans below and above it, narrowing toward the lower end as myopia increases — the two results are not in conflict, they answer different questions about the same underlying trade-off, as established in Chapter 5.

Part VI

Conclusion

Chapter 11

Conclusion

11.1 What is established

- The single-hole wave-optics pipeline (Chapter 3, Chapter 6) is correctly built: the FFT-at-retina failure mode is identified and fixed, the fix is validated against the exact Fraunhofer and on-axis Fresnel-oscillation closed forms, and the deprecated shortcuts it replaces (geometric-blur addition, RSS combination) have their errors directly measured rather than merely argued away.
- The optimisation objective is visual acuity, not blur radius (Chapter 5), and the correction from one to the other is itself a measured, reported result — including the finding that the cited brightness term is inert at every design actually selected.
- The array's ghost-safety does not depend on the one uncalibrated parameter inherited from prior literature; it is structurally non-binding by the pitch-equals-pupil-diameter architecture, verified directly from generated coordinates (Chapter 4).
- ASM measurably outperforms RSS against real photographic data, though the correct, narrower reading of that comparison — not an unqualified confirmation of ASM — is reported in Chapter 10.
- An independent optical-bench sweep (ExP3) confirms the pinholes are doing real optical work relative to an unrestricted control, while reporting an aperture-slope ordering that does not match the geometric-blur prediction, plainly and without a forced explanation.

11.2 What is not yet established

- The acuity predictor has one external validation attempt on record, against Wang (2009), and it failed: negative R^2 at every M tested, no statistically significant relationship detected. The leading suspects are an uncalibrated formula and a small, non-monotonic clinical dataset, not necessarily the underlying optics — but this remains unresolved, not merely caveated.
- The array's fusion threshold $c_{\min}$ is argued non-binding by geometry, not by direct measurement; the two-pinhole calibration experiment Shen's own authors name as the correct test has not been run, for lack of resources.
- No calibrated logMAR acuity measurement has yet been taken against the fabricated array (Chapter 9); the protocol is specified (Chapter 8) and the model's own sharpest falsifiable prediction — flatness of pinhole-condition acuity across myopia levels (Chapter 7) — is designed to be tested by it.

- Transmission at high myopia is genuinely low (1% at $M = 8\,\mathrm{D}$), a direct and undisguised cost of the ghost-free sparse architecture, not a free result.

11.3 Immediate next steps, in priority order

1. Run the logMAR acuity protocol (Chapter 8) against the fabricated array, testing specifically for the flatness signature (Chapter 7) across subjects of varying myopia.
2. Complete post-etch verification of the fabricated part and insert the measured values into Chapter 9.
3. Run the $c_{\min}$ sensitivity sweep specified in Chapter 4 — pure compute, no additional apparatus required — to convert the geometric non-binding argument into a demonstrated one.
4. State the acuity-model validation status (Chapter 10) explicitly and first, in any presentation of this work, rather than let it be discovered.

This document reports every result at the standard set by this project's own earlier experimental work: a finding that fails a check is reported as failing, a formula whose error was measured reports the measured number rather than a qualitative reassurance, and a result still open is left open rather than rounded off. That discipline is, on the evidence of this document, the project's most defensible property, and maintaining it through presentation and defence matters more than any single number inside it.

Appendix A

Formula Quick-Reference

Table A.1: Every working formula in this handbook, with status.

Quantity	Formula	Status
Myopic axial length	$L_{\text{ax}} = n'/(P_0 - M)$	live
Emmetropic axial length	$L_0 = n'/P_0 = 22.267\,\text{mm}$	live
ASM propagator	$H = \exp(ikz\sqrt{1 - \lambda^2(f_x^2 + f_y^2)})$	live, used twice
Fresnel number	$N_F = R_{\text{hole}}^2/(\lambda z)$	live (regime gate)
Airy intensity	$I = I_0(2J_1(x)/x)^2$	validation only
Encircled energy	$EE = 1 - J_0^2(x) - J_1^2(x)$	validation only
On-axis Fresnel oscillation	$I(0, z) = 4I_0 \sin^2(\pi N_F/2)$	live (cross-check)
Relative illuminance	$E_{\text{rel}} = (\min(a_{\text{hole}}, D_{\text{pupil}})/D_{\text{pupil}})^2$	live
Hole–pupil overlap test	$d - R_h < R_p$	live
Gaze pupil shift	$\Delta = R_{\text{rot}} \sin\varphi$	live
Ghost separation	$\Delta r = sM/(P_0 - M)$	live, $R^2 = 0.995$
Packing density area	$A_{\text{hull}} + P_{\text{hull}}R_h + \pi R_h^2$	live
Acuity (recommended)	$\text{MAR} = (30/f_{\text{MTF10}})\text{VSOTF}^{-0.1}$	live
Brightness correction	$0.060 \log_{10}(300/\text{Td})$ below knee	live, cited
Undercut compensation	$d_{\text{mask}} = d_{\text{target}} - C$, $C \in [0.5t, 1.5t]$	live, uncalibrated bracket
Geometric blur (old)	$r_{\text{geo}} = \frac{a_{\text{stop}}}{2}\frac{\lvert v_r - v_f\rvert}{v_f}$	**deprecated**
RSS combination	$\sqrt{r_{\text{geo}}^2 + r_{\text{diff}}^2}$	**deprecated, 64.8% mean error**
Quadrature a_{eff}	$\sqrt{a_{\text{hole}}^2 + (2.44\lambda d_{\text{gap}}/a_{\text{hole}})^2}$	**deprecated**

Appendix B

Constants Used Throughout

Symbol	Value	Meaning
P_0	60.0 D	emmetropic eye power, held fixed
n'	1.336	vitreous refractive index
D_{pupil}	4000 μm	default pupil diameter
R_{rot}	13–14 mm	eye rotation radius
λ	550 nm	default design wavelength
d_{gap}	12 mm	spectacle vertex distance
$c_{\min}$	55 μm	Shen fusion floor — inherited, non-binding for this design
t	50 μm	brass shim thickness
Scene luminance	10 000 cd/m^2	bright-daylight default; see Ch. 5, Ch. 9, Ch. 10 for the indoor-measurement confound this creates
Weber knee	300 Td	Van Nes & Bouman (1967)
Brightness slope	0.060 logMAR/decade	Eidbury et al. (2016)
Cycles/deg at 20/20	30.0	logMAR scale anchor
Calibration square	10.0 mm	print-scale verification on every mask